

# RCP Bearing Inspection & Replacement (PM-7B)

Preventive Maintenance Procedure 0PMP-RCS-7B · Maintenance / Mechanical

|                 |                                      |                |                          |
|-----------------|--------------------------------------|----------------|--------------------------|
| DOCUMENT NUMBER | 0PMP-RCS-7B                          | CLASSIFICATION | INTERNAL                 |
| TITLE           | RCP Bearing Inspection & Replacement | REVISION       | Rev 2                    |
| EFFECTIVE DATE  | 2024-09-01                           | OWNER          | Maintenance / Mechanical |

## 1.0 Scope

This procedure governs the scheduled inspection of RCP thrust and journal bearings on Westinghouse Model-93A reactor coolant pumps at intervals of **730 days**, in conformance with Tech Spec 3.4.5 and ASME Section XI.

## 2.0 Prerequisites

- Unit shall be in MODE 5 (refueling) or MODE 6 (defueled) prior to procedure entry.
- RCP breaker shall be open with two-person LOTO verification per OSHA 1910.147.
- Bearing oil sample collection equipment available (3 × 50ml clean sample bottles).
- Calibrated vibration data collector (calibration current within 30 days).

## 3.0 Personnel Qualifications

Lead engineer shall hold ASME Section XI Inspector certification. Technicians shall hold Mechanical Maintenance II minimum.

## 4.0 Acceptance Criteria

**Bearing axial vibration shall be  $\leq 0.20$  in/s post-maintenance; thrust play shall be  $0.020 \pm 0.005$  in. Out-of-spec readings require Engineering Disposition.**

## 5.0 Procedure Steps

- 5.1 Verify LOTO applied. Two-person verification documented on attached sign-off form.
- 5.2 Remove upper bearing housing access cover. Inspect for cracks, deformation.
- 5.3 Visually inspect thrust bearing surfaces. Acceptance: no scoring, pitting, or unusual wear.
- 5.4 Collect vibration baseline data — axial and radial channels — with calibrated meter.
- 5.5 Collect three 50ml bearing oil samples for chemistry analysis. Label with sample ID format BL-YYYY-MMDD-<eq>-<n>.
- 5.6 Visually inspect mechanical seal cavity. Acceptance: no leak signature, no oil staining.
- 5.7 Reinstall bearing housing cover. Torque to **95 ft-lbs  $\pm$  5** per Westinghouse specification.

5.8 Remove LOTO and restore RCP breaker.

5.9 Perform post-maintenance functional test: 50% rated speed × 30 min. Acceptable if vibration meets §4.0 criteria.

## 6.0 Records

Completed work package shall be filed in eDM under WO-YYYY-NNNNN. Bearing oil sample chemistry results shall be filed in chemistry record CR-RCS-<equipment>-<date>. Records retained per 10 CFR 50.71 and STP-001.

## 7.0 References

- Tech Spec 3.4.5 — Reactor Coolant System Operability
- ASME Section XI — Rules for In-Service Inspection
- Westinghouse RCP Maintenance Manual, Section 7.4
- OSHA 1910.147 — Lockout/Tagout
- STP-OP-2204 — Reactor Coolant Pump Surveillance & Lockout